Partial order of signed graph homomorphisms

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Abstract

A signed graph (G, σ) is a graph G together with an assignment σ which assigns to each edge of G a sign, either positive or negative. The sign of a closed walk in (G, σ) is the product of the signs of its edges (considering multiplicity). Considering signs of closed walks as one of the key structures of signed graphs, a homomorphism of a signed graph (G, σ) to a signed graph (H, π) is defined to be a mapping of the vertices and edges of G to the vertices and edges of H (respectively) which preserves incidences, adjacencies and signs of closed walks. This is a recently defined notion that is closely related to sign-preserving homomorphisms of signed graphs (or, equivalently, to homomorphisms of 2-edge-colored graphs) which helps, in particular, to establish a stronger connection between the theories of coloring and homomorphisms of graphs and the minor theory of graphs.

When there exists a homomorphism of (G, σ) to (H, π), one may write $(G, \sigma) \to (H, \pi)$, thus extending the graph homomorphism order to an order on the class of signed graphs. In this work, studying this order, we prove that, when restricted to the class of cores, this order forms a lattice. That is to say that for each pair (G_1, σ_1) and (G_1, σ_1) of signed graphs, both their meet and joint exist.

1 Introduction

A signed graph (G, σ) is a graph G together with an assignment σ which assigns to each edge either a positive or a negative sign. The mapping σ can be given by the set E^- of negative edges, in which case we may write (G, E^-) instead. Given a signed graph (G, σ), the *signature* of G normally denotes the set of its negative edges, but it may also refer to the function σ in an equivalent way.

For the purpose of this work, graphs are finite and are allowed to have loops and multi-edges. However, multi-edges of a same sign will be irrelevant to our work and will thus not be considered. A pair of parallel edges of different signs (together with its endpoints) is referred to as a *digon*.

Two key notions in the study of signed graphs are *signs of substructures*, which is defined to be the product of the signs of the edges in a given structure, and *switching*, which is to change the sign of each edge in a subset of edges. In most studies, including this one, these two notions are restricted to two types of substructures which are closely related: sign will be defined for cycles, or more generally for closed walks, and switching can be done on an edge cut, noting that these two structures are rather dual. They are formally defined as follows.

Definition 1 (Sign of cycles and closed walks, switching, switching equivalence)**.** Given a signed graph (G, σ) and a closed walk W of G, the *sign of* W, denoted $\sigma(W)$, is the product of the signs of

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the edges of W, allowing multiplicities. A switch of (G, σ) on an edge cut (X, Y) of G is to change the signs of all edges in (X, Y). Furthermore, (G, σ') is said to be *switching equivalent* to (G, σ), if (G, σ') is a switch of (G, σ).

The following observations directly follow from these definitions.

Observation 2. *Given a signed graph (G, σ), the sign of every closed walk of G is invariant under the switch operation.*

Observation 3. *Given two switch equivalent signed graphs (G, σ) and (G, σ'), the symmetric difference of the two signatures is an edge cut of G.*

Switching equivalence partitions the set of all signatures on a graph G into equivalence classes where all members of a class partition the set of cycles (or closed walks) into two parts, the negative ones and the positive one. The following theorem of Zaslavsky shows that this partition uniquely determines the equivalence class to which a signature belongs.

Theorem 4. *[7] Two signed graphs (G, σ_1) and (G, σ_2) are switching equivalent if and only if they have the same set of negative cycles.*

It should be noted here that not every partition of cycles or closed walks of a graph G into two parts would correspond to a set of positive and negative cycles of a signature on G. A classification of such partitions, using the notion of *exclusive 3-walk property*, can be found in [5] and [6].

One should also note that a signed graph (G, σ) could be simply viewed as a 2-edge-colored graph, usually associating positive edges with color blue (or with solid lines) and negative edges with color red (or with dashed lines). With such a view, all results on 2-edge-colored graphs indeed hold for signed graphs. However, the notion of switch, which is based on the basic mathematical relation between $+$ and $-$ (as opposed to colors red and blue which are not related), adds an intriguing level of complexity which gives birth to a number of beautiful theories.

A particular example is the notion of *homomorphisms* of signed graphs. While it remains strongly related to homomorphisms of 2-edge-colored graphs, it becomes of higher interest in the study of a number of conjectures, specially those in relation with the theory of graph minors, such as the four-color theorem and its possible extensions.

In this paper, we study the homomorphism order on signed graphs. Because of the strong relation with homomorphisms of 2-edge-colored graphs, we will also consider the homomorphism order on 2-edge-colored graphs, but, to emphasize on the connection, and toward a uniform presentation, we will study homomorphisms of 2-edge-colored graphs under the umbrella of edge-sign-preserving homomorphisms of signed graphs.

We continue with proper definitions of the two homomorphism relations and their connections in the next section.

2 Homomorphisms of signed graphs

Considering signs of cycles and closed walks as one of the key structures of a signed graph, the following is the natural definition of homomorphisms of signed graphs.

Definition 5 (Homomorphism of signed graphs [5])**.** Given two signed graphs (G, σ) and (H, π), a homomorphism of (G, σ) to (H, π) is a mapping ϕ which maps $V(G)$ to $V(H)$, $E(G)$ to $E(H)$, in such a way that adjacencies, incidences and signs of closed walks are preserved. When such a homomorphism exists, we write $(G, \sigma) \to (H, \pi)$.

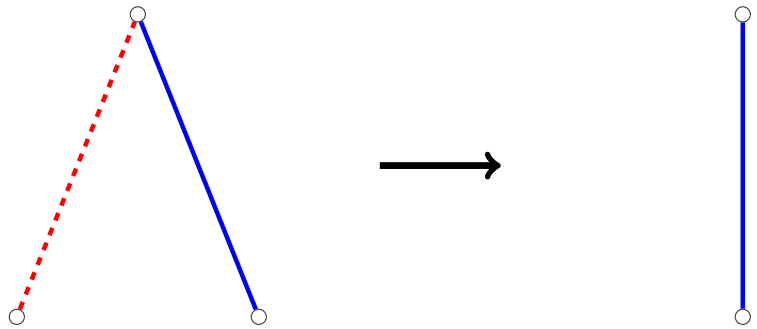

Figure 1: An example of a homomorphism of signed graphs

Observe that, in a mapping of (G, σ) to (H, π), when vertices x and y of an edge $e = xy$ of G are mapped to the two ends of a digon in (H, π), or to the same vertex which has both a negative and a positive loop, then one must precise to which of the two possible edges e is mapped to. In all other cases, specially when H has no multi-edge, the edge mapping is induced by the vertex mapping and one may simply refer to the vertex mapping.

Observe also that in this definition of homomorphism, a mapping ϕ of (G, σ) to (H, π) may map a positive edge to a negative one and vice versa. An easy example of such a homomorphism is given in Figure 2. However, there are strong connections between our definition and homomorphisms that preserve the signs. Two key connections will be discussed later. For more on that point, we refer to [5].

Definition 6 (Edge-sign-preserving homomorphism of signed graphs)**.** Given two signed graphs (G, σ) and (H, π), an *edge-sign-preserving homomorphism* of (G, σ) to (H, π) is a mapping ϕ which maps $V(G)$ to $V(H)$, $E(G)$ to $E(H)$, in such a way that adjacencies, incidences and signs of edges are preserved. When such a homomorphism exists, we write $(G, \sigma) \xrightarrow{s.p.} (H, \pi)$.

As in this work we only consider signs for edges, we may simply write sign-preserving homomorphism in place of edge-sign-preserving homomorphism.

With respect to each of these two definitions, we can associate a notion of core: the *core* of a signed graph (G, σ) is the minimal subgraph of (G, σ) to which (G, σ) admits a homomorphism. The fact that this notion is well defined follows from the fact that any two such minimal subgraphs are necessarily isomorphic, as shown in [4]. In each case, isomorphism is defined, according to the corresponding notion of homomorphism, as a one-to-one and onto homomorphism. For example, the signed graph on the left of Figure 2 is switching equivalent to the signed graph obtained by switching its negative edge to a positive one, and its core is the graph on the right of the figure. Furthermore, as a 2-edge-colored graph or with respect to sign-preserving homomorphisms, the graph on the right is its own core as it does not map to any of its proper subgraphs.

A signed graph (G, σ) whose core (according to the two notion of homomorphism or of sign-preserving homomorphism) is (G, σ) itself is called a *core*. To distinguish these two notions, we will refer to the latter one as an *sp-core* (of (G, σ)). Observe that the core of (G, σ) must be a subgraph of an sp-core of (G, σ) and, in some case, the two might be identical (consider for example a negative cycle).

It is easily observed that both relations $\rightarrow$ and $\xrightarrow{s.p.}$ are reflexive and associative. However, considered on the class of all signed graphs, none of them is antisymmetric. To form a partial order from either of them, we must restrict ourselves to the class of cores or sp-cores, or, equivalently, we may put all homomorphically equivalent signed graphs into a same class, presented by any of its members, preferably by its smallest element which is the (unique) core. (Note here that this might be a difficult task to find the core of a given class of signed graphs.) The main focus of this paper is the study of these partial orders.

It is a well known fact that the homomorphism order of graphs restricted to the class of cores is a lattice, that is to say that any pair of elements admits a joint and a meet. The *joint* of two given elements is simply (the core of) their disjoint union, and their *meet* is given by a graph product defined below. This product is known under various names such as *categorical product*, *direct product*, *tensor product*, *Hedetniemi product*, etc. We refer to [2] for more details.

Definition 7 (Categorical product of graphs)**.** Given two graphs G and H, the *direct product* of G and H, denoted $G \times H$, is the graph with vertex set $V(G) \times V(H)$ where a vertex (x, u) is adjacent to a vertex (y, v) if and only if x is adjacent to y in G and u is adjacent to v in H.

This product has been naturally extended to edge-colored and directed graphs, using the fact that the homomorphism order induced the corresponding class of cores is a lattice.

Definition 8 (Sp-direct product of signed graphs)**.** Given two signed graphs (G, σ) and (H, π), the *sp-direct product* of (G, σ) and (H, π), denoted $(G, \sigma) \overset{sp}{\times} (H, \pi)$, is the signed graph with vertex set $V(G) \times V(H)$, where a vertex (x, u) is adjacent to a vertex (y, v) with a positive edge if and only if x is adjacent to y in (G, σ) with a positive edge, and u is adjacent to v in (H, π) with a positive edge. The set of negative edges is defined analogously.

We will show in the next section that the homomorphism order induced on the class of signed graphs is also a lattice. The product of two signed graphs will be defined in two ways. A theoretically simpler one is through a strong connection with sign-preserving homomorphisms, which is described below. We refer to [5] and references therein for more on this connection.

The first of these connections comes from the following theorem which was taken as the definition of homomorphisms of signed graphs in [4].

Theorem 9 ([5])**.** *Given two signed graphs (G, σ) and (H, π), there exists a homomorphism of (G, σ) to (H, π) if and only if there is a signed graph (G, σ'), switching equivalent to (G, σ), and a sign-preserving homomorphism of (G, σ') to (H, π).*

The second connection we would like to mention is based on the notion of double switch graph which was introduced in [1] within a larger context. Our terminology is taken from [5].

Definition 10 (Double switch graph of a signed graph)**.** Given a signed graph (G, σ), the *double switch graph of* (G, σ), denoted $DSG(G, \sigma)$, is the signed graph whose set of vertices is two disjoint copies V^+ and V^- of the vertices of G, where each copy induces a copy of (G, σ) and, furthermore, where two vertices $x^+ \in V^+$ and $y^- \in V^-$ are adjacent with a positive edge (respectively, with a negative edge) if and only if the vertices x and y are adjacent in (G, σ) with a negative edge (respectively, with a positive edge).

Observe that if (G, σ') is obtained from (G, σ) by switching at a set X of vertices, then the corresponding sets $X^- \subset V^-$ and $(V \setminus X)^+ \subset V^+$ induce a copy of (G, σ'), with their complement inducing another copy of the same signed graph. This explains why this construction is called the double switch graph.

The following result is a special case of a general theorem given in [1].

Theorem 11. *Given two signed graphs (G, σ) and (H, π), $(G, \sigma) \to (H, \pi)$ if and only if $(G, \sigma) \xrightarrow{s.p.} DSG(H, \pi)$.*

In practice, we would normally take the conditions of Theorem 9 to define a mapping that is a homomorphism of (G, σ) to (H, π). Therefore, such a mapping f is composed of three parts, say $f = (f_1, f_2, f_3)$, where f_1 decides for each vertex x whether there would be a switch at x or not, f_2 decides to which vertex of H the vertex x is mapped, and f_3 determines the images of the edges. As mentioned before, when there is no multi-edge, f_3 is completely determined by f_2, and f could be

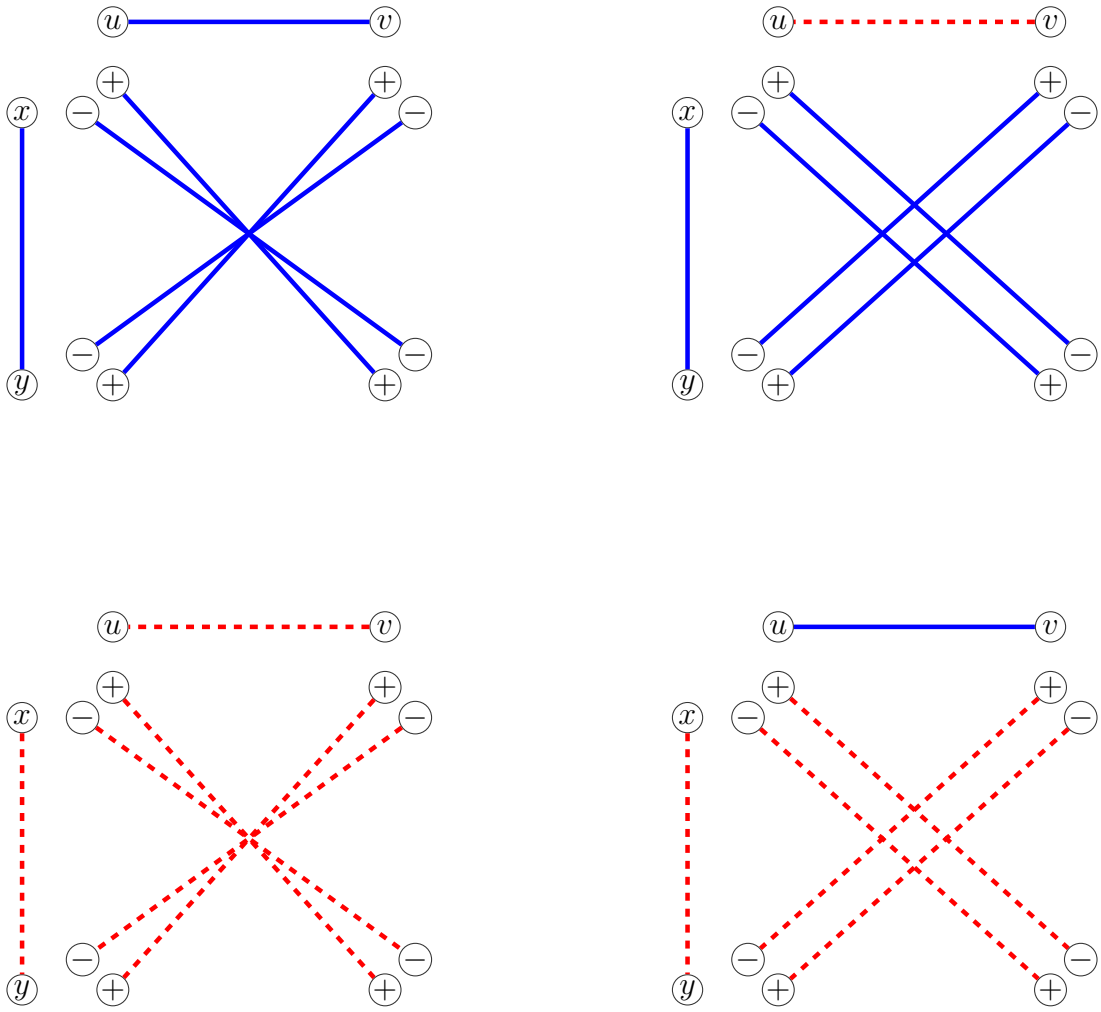

Figure 2: Possible products of two edges

simply presented by f_1 and f_2. Moreover, the essence of Theorem 11 is that if f_1 applies a switch at some vertex x, then x would be mapped to a vertex in the V^- part of $DSG(H, \pi)$, otherwise x would be mapped to a vertex in the V^+ part. Thus, using $DSG(H, \pi)$, one may simplify the mapping into a single function at the cost of doubling the number of possible images.

3 The homomorphism order of signed graphs

Our goal in this section is to prove that the homomorphism order on signed graphs induced on the class of cores is a lattice. That is to say, for any pair of signed graphs (G, σ) and (H, π), both their joint and meet exist.

Definition 12 (Meet and categorical product of two signed graphs)**.** Given two signed graphs (G, σ) and (H, π), the categorical product of (G, σ) and (H, π), denoted $(G, \sigma) \times (H, \pi)$, is the signed graph (F, ς) defined as follows.

1. $V(F) = \{+, -\} \times V(G) \times V(H)$.

2. To have an edge between (i, x, u) (j, y, v), first of all we must have an edge xy in G and an edge uv in H. In such a case, the corresponding eight vertices (respectively, four vertices if either $x = y$ and $u = v$, or two vertices if both $x = y$ and $u = v$) will induce a matching of order four (respectively, two, one), all of its edges having the same sign as $\sigma(xy)$. The choice of these

edges then depends on whether uv has the same sign as xy or not, and is described as follows (see Figure 2).

(a) If $\sigma(xy) = \pi(uv)$, then (i, x, u) is adjacent to (i, y, v) and (i, x, v) is adjacent to (i, y, u) for every $i \in \{+, -\}$ (left side of Figure 2).

(b) If $\sigma(xy) = -\pi(uv)$, then (i, x, u) is adjacent to $(-i, y, v)$ and (i, x, v) is adjacent to $(-i, y, u)$ for every $i \in \{+, -\}$ (right side of Figure 2).

We will show that not only $(G, \sigma) \times (H, \pi)$ is the meet of (G, σ) and (H, π) in the homomorphism order, but, moreover, that it is the categorical product of (G, σ) and (H, π), that is, (i) there are two natural projections P_G and P_H that map this product graph to (G, σ) and (H, π), respectively, and (ii) for every signed graph (C, η), if $(C, \eta) \to (G, \sigma)$ with a mapping f_G and $(C, \eta) \to (H, \pi)$ with a mapping f_H, then $(C, \eta) \to (G, \sigma) \times (H, \pi)$ with a mapping f such that $f \circ f_G = P_G$ and $f \circ f_H = P_H$.

Theorem 13. *Given two signed graphs (G, σ) and (H, π), the signed graph $(G, \sigma) \times (H, \pi)$ is the categorical product of (G, σ) and (H, π).*

Proof. We must first present the natural projections P_G and P_H of $(G, \sigma) \times (H, \pi)$ onto (G, σ) and onto (H, π), respectively. The projection P_G maps each vertex (i, x, v) to the vertex x of (G, σ) without applying a switching. It is straightforward to check that P_G is a homomorphism of $(G, \sigma) \times (H, \pi)$ to (G, σ). The projection P_H first applies a switch at all vertices of the form $(-, x, v)$, and then maps each vertex (i, x, v) to the vertex v of (H, π). It is again easy to verify that P_H is a homomorphism of $(G, \sigma) \times (H, \pi)$ to (H, π).

We may now consider a signed graph (C, η) and assume that f_G and f_H are homomorphisms of (C, η) to (G, σ) and (H, π), respectively. We would like to define a natural mapping f of (C, η) to the product $(G, \sigma) \times (H, \pi)$ using f_G and f_H. Recall that a homomorphism f of signed graphs consists of three components, f_1, f_2 and f_3, the first one deciding at each vertex if a switching must be done, the second one being a vertex mapping and the third one being an edge mapping.

To define the mapping f, we choose f_1 to be identical to the first component of f_G, that is to say, f_1 applies a switch at a vertex z of (C, η) if and only if f_G applies a switch at z. For the component f_2, if f_H applies a switch at a vertex z of (C, η), then f_2 maps z to $(-, f_G(z), f_H(z))$, otherwise f_2 maps z to $(+, f_G(z), f_H(z))$.

Finally, we must determine the component f_3, that is the edge mapping. Let e an edge of (C, η) with endpoints z and z'. Let i be the sign of e after applying the possible switches given by f_G at vertices z and z'. Observe that there must be an edge with sign i between $f(z)$ and $f(z')$. We then set $f_3(e)$ to be this edge. It is then straightforward to check that f is a homomorphism of (C, η) to $(G, \sigma) \times (H, \pi)$ and, furthermore, that $P_G \circ f = f_G$ and $P_H \circ f = f_H$. □

The product graph above defined has order $2 \times |V(G)| \times |V(H)|$. While even the product of two cores is not always a core, for some choices of cores, the product graph could be a core (see Section 4), and thus, as a general construction that works for all pairs of signed graphs, this is the most optimal description of the product. However, from a theoretical point of view, it is easier to describe a homomorphically equivalent signed graph which has twice this order, that is a signed graph on $4 \times |V(G)| \times |V(H)|$ vertices. To describe this construction, we must introduce a few more concepts.

Given a signed graph (G, σ) and a vertex v of G, we denote by $N^+(v)$ (respectively, by $N^-(v)$) the set of vertices of G that are adjacent to v by a positive edge (respectively, by a negative edge), allowing $N^+(v)$ (respectively, $N^-(v)$) to contain v only if there is a positive loop at v (respectively, a negative loop at v). Two vertices u and v of a signed graph are said to be *twins* if and only if $N^+(v) = N^+(u)$ and $N^-(v) = N^-(u)$. On one other hand, two vertices u and v are *antitwins* if $N^+(v) = N^-(u)$ and $N^-(v) = N^+(u)$, or, equivalently, if after a switch (only) at u, v and u are

twins in the switching equivalent signed graph (G, σ'). Observe that our definition of twins deals with loops as well. Indeed, if u is an antitwin of v in (G, σ) and u has, say, a positive loop, then u must be adjacent to v by a negative edge, and v must also have a positive loop.

Clearly, the core of (G, σ), with respect to either of the two homomorphism notions we have considered, cannot contain any pair of twin vertices. The core of (G, σ) with respect to sign-preserving homomorphisms may contain a pair of antitwin vertices. However, the core with respect to homomorphisms cannot contain any pair of antitwin vertices either, as otherwise one may do a switching at one of them and then identify them. As two antitwins of a same vertex must themselves be twins, we get that each vertex can have at most one antitwin in the sp-core of (G, σ). Observe also that in $DSG(G, \sigma)$, the two copies in V^+ and V^- of any vertex x are antitwins. The converse is also true in the following sense.

Theorem 14. *A signed graph (H, π) is the double switch graph of a signed graph (G, σ) if and only if each vertex of (H, π) has a unique antitwin in (H, π).*

Proof. One direction directly follows from the definition of $DSG(G, \sigma)$. For the other direction, simply delete one vertex from each pair of antitwin vertices in (H, π) and let (G, σ) be the resulting signed graph. It is clear that different choices will lead to switching equivalent copies of (G, σ). □

The direct product of two signed graphs with respect to sign-preserving homomorphisms is their direct product when considered as 2-edge-colored graphs, defined as a natural extension of the direct product of graphs. More precisely, it is defined as follows.

Definition 15 (Direct product with respect to sign-preserving homomorphisms)**.** Given two signed graphs (G, σ) and (H, π), the *direct product of (G, σ) and (H, π) with respect to sign-preserving homomorphisms*, denoted $(G, \sigma) \overset{sp}{\times} (H, \pi)$, is the signed graph with vertex set $V(G) \times V(H)$, where vertices (x, u) and (y, v) are adjacent with a positive edge if and only if x and y are adjacent with a positive edge in (G, σ) and$u$ and v are adjacent with a positive edge in (H, π). The set of negative edges is defined analogously.

We may now combine Definition 15 with Theorem 14 to redefine the product of two signed graphs (G, σ) and (H, π) with respect to homomorphisms of signed graphs (see Definition 12), as the signed graph (F, η) whose double signed graph is the product $DSG(G, \sigma) \overset{sp}{\times} DSG(H, \pi)$ (note that this product graph has order $4 \times |V(G)| \times |V(H)|$). This is well defined since we have the following result.

Theorem 16. *Given two signed graphs (G, σ) and (H, π), the set of vertices of the direct product $DSG(G, \sigma) \overset{sp}{\times} DSG(H, \pi)$ can be partitioned into pairs of antitwins.*

Proof. Recall that, for each vertex x of G, there are two vertices x^+ and x^- in $DSG(G, \sigma)$. Thus, for each pair of vertices (x, u) with $x \in V(G)$ and $u \in V(H)$, we have four vertices (x^+, u^+), (x^+, u^-), (x^-, u^+) and (x^-, u^-) in $DSG(G, \sigma) \overset{sp}{\times} DSG(H, \pi)$. Furthermore, if any vertex, say (y^+, v^+), is adjacent to one of these four vertices, say (x^+, u^+), with a positive edge, that would mean that xy is a positive edge of (G, σ) and uv is a positive edge of (H, π). Thus, x^-y^+ is a negative edge of $DSG(G, \sigma)$ and u^-v^+ is a negative edge of $DSG(H, \pi)$, which gives that (x^-, u^-) is the antitwin of (x^+, u^+) in $DSG(G, \sigma) \overset{sp}{\times} DSG(H, \pi)$. Similarly, (x^-, u^+) and (x^+, u^-) form a pair of antitwin vertices in $DSG(G, \sigma) \overset{sp}{\times} DSG(H, \pi)$.

All other cases being similar, this concludes the proof. □

In other words, the product (F, ς) of (G, σ) and (H, π) satisfies

$$DSG(F, \varsigma) = DSG(G, \sigma) \overset{sp}{\times} DSG(H, \pi,)$$

where (x^+, u^+) is taken as the vertex $(+, x, u)$ of (F, ς), which represents the pair (x^+, u^+) and (x^-, u^-) of antitwin vertices in $DSG(G, \sigma) \overset{sp}{\times} DSG(H, \pi)$, and (x^+, u^-) is taken as the vertex $(-, x, u)$, which represents the pair (x^-, u^+) and (x^+, u^-). Hence, the given signature ς is a function of σ, π, and their order in the product. However, the class of switching equivalent signatures to which ς belongs is only a function of σ and π, and is independent of the order in which we multiply the two signed graphs. In other words, $(G, \sigma) \times (H, \pi)$ and $(H, \pi) \times (G, \sigma)$ are switch equivalent. Furthermore, if (G, σ') is switch equivalent to (G, σ) and (H, π') is switch equivalent to (H, π), then $(G, \sigma') \times (H, \pi')$ is switch equivalent to (F, ς).

4 Remarks

In this work, given two signed graphs (G, σ) and (H, π), we have defined their categorical product $(G, \sigma) \times (H, \pi)$. We then defined the homomorphism order of signed graphs, whose elements are the classes of signed graphs, elements of each class being all the signed graph which are pairwise homomorphically equivalent, two signed graphs (G, σ) and (H, π) being homomorphically equivalent if and only if $(G, \sigma) \to (H, \pi)$ and $(H, \pi) \to (G, \sigma)$.

We then showed that the homomorphism order of signed graphs is a lattice, by showing that the categorical product of (G, σ) and (H, π) is their meet in the order. We gave two constructions, one with order $2 \times |V(G)| \times |V(H)|$ and another, which was theoretically simpler, with order $4 \times |V(G)| \times |V(H)|$. We showed furthermore that this product is the categorical product of the two signed graphs.

The two examples below will show that (1) the core of the product of a given pair of signed graphs could be of order smaller than the order of the two given signed graphs, and (2) as a general construction that works for all pairs of signed graphs, $2 \times |V(G)| \times |V(H)|$ is the best order one may work with.

Given an integer k, $k \neq 0$, let C_k be a signed graph on the cycle of length $|k|$ (i.e., $C_{|k|}$) with an assignment of signs to the edges in such a way that the sign of the resulting cycle is k, that is to say, C_k is a negative cycle if $k < 0$ and a positive cycle otherwise. In particular, C_{-1} is a negative loop and C_{+1} is the positive loop. The product of two cycles C_p and C_q can then be determined as follows.

1. If p and q have the same parity and the same sign, then, assuming $|p| \leq |q|$, $C_p \times C_q = C_q$. That is simply because $C_q \to C_p$, and thus the signed graphs admitting a homomorphism to both C_p and C_q are those admitting a homomorphism to C_q.

2. If either the parity or the sign of p and q are not the same, then the product of C_p and C_q is simply $(K_2, +)$, which is homomorphically equivalent to $(K_2, -)$. To see that, let us denote by $g_{ij}(G, \sigma)$, $ij \in \mathbb{Z}_2^2$, the smallest length of a closed walk in (G, σ) whose sign is positive if $i = 0$ and negative if $i = 1$, and whose length is even if $j = 0$ and odd if $j = 1$ (see [5] for more details on this notion). If no such closed walk exists in (G, σ), we let $g_{ij}(G, \sigma) = \infty$. Observe now that a signed graph (G, σ) which maps to C_p has $g_{ij}(G, \sigma) = \infty$ for at least two choices of $ij \in \mathbb{Z}_2^2$, where these two choices depends on the parity and sign of p. As the same holds for q, and since p and q cannot lead to the same choices, a signed graph (G, σ) which maps to both of them must have $g_{ij}(G, \sigma) = \infty$ for at least three choices of $ij \in \mathbb{Z}_2^2$. Therefore, the only possible closed walks in (G, σ) are closed walks of type 00. Thus, on the one hand $C_{-1} \times C_{+1}$ has $2 \times |V(C_{-1})| \times |V(C_{+1})|$ vertices, and on the other hand, for large values of $|p|$ and $|q|$ where either the parity of the sign of p and q are not the same, the core of the product $C_p \times C_q$ is of order 2.

Finally, it is worth noting here that similar constructions can be given, leading to similar results, for oriented graphs with respect to pushable homomorphisms, as introduced by Klostermeyer and MacGillivray in [3].

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